

CLINICAL

FINDINGS:

CUTANEOUS LESIONS

The clinical presentation of Cushing syndrome may be subtle. Skin findings are often present, and when carefully correlated with systemic complaints, they can lead to an early diagnosis. Common skin manifestations include easy bruisability and skin thinning. Mild hypertrichosis, typically of the lanugo type, and acne may also occur. Frank hirsutism and severe acne suggest the possibility of a concurrent androgen-secreting tumor.

The development of multiple violaceous striae wider than 1 cm on the abdomen or proximal extremities is highly suggestive of Cushing syndrome. The width and violaceous color of these striae help differentiate them from the narrower, hypopigmented striae commonly seen in pregnancy or with rapid weight gain.

Relatively dramatic changes in fat distribution often accompany generalized weight gain. There is increased central adiposity with thinning of the extremities. Central obesity, along with facial fat accumulation, leads to a rounded facial appearance known as **moon facies**, often accompanied by facial erythema and telangiectasias. Additional fat deposition occurs in the dorsocervical region (buffalo hump) and supraclavicular fossae. Men may develop **gynecomastia**.

Although uncommon, **hyperpigmentation** can occur in Cushing syndrome, particularly when there is marked overproduction of ACTH—most often from an ectopic source, but sometimes from the pituitary. Both ACTH and melanocyte-stimulating hormone (MSH), derived from proopiomelanocortin (POMC), contribute to increased melanin production, similar to what is seen in Addison disease.

Acanthosis nigricans may be present due to insulin resistance associated with Cushing syndrome, which often leads to diabetes mellitus. The

immunosuppressive effects of excess glucocorticoids predispose patients to increased **dermatophyte and candidal infections** of the skin and nails.

Women may present with **irregular menses or amenorrhea**. Other systemic features include **hypertension, osteoporosis, muscle wasting, and proximal muscle weakness**.

TABLE 152-11

Approach to the Patient with Cushing Syndrome

Cutaneous findings in Cushing syndrome

- Increased central adiposity (moon facies and buffalo hump) with thinning of the extremities
- Skin thinning and easy bruisability
- Violaceous striae
- Acanthosis nigricans
- Increased dermatophyte and candidal skin and nail infections
- Rarely, hyperpigmentation

Related features

- Diabetes
- Hypertension
- Osteoporosis
- Irregular menses

Diagnosis

- Measurement of cortisol in serum and urine

- Failure to suppress cortisol production with dexamethasone suppression tests

Management

- Appropriate endocrine and surgical consultation
- Surgical removal of pituitary or adrenal adenoma
- Adrenal enzyme inhibitors
- Radiation therapy

LABORATORY TESTS

Diagnostic confirmation of Cushing syndrome requires demonstration of excess cortisol production. Cortisol levels exhibit diurnal variation, typically peaking in the early morning. Therefore, serum cortisol should be measured at 8 A.M.

The average daily cortisol production can be assessed via **24-hour urinary free cortisol** measurement. Any abnormal result should be confirmed with repeat testing (two to three times).

If urinary or serum cortisol levels are elevated, the diagnosis is supported by **failure to suppress** cortisol during a **low-dose dexamethasone suppression test** (e.g., 1 mg overnight or 2 mg over 48 hours).

Further testing, such as the **high-dose dexamethasone suppression test**, helps differentiate pituitary-dependent Cushing disease from ectopic ACTH production or adrenal tumors.

It is important to note that **hypercortisolism** can be present in the absence of true Cushing syndrome (e.g., in pseudo-Cushing states such as obesity, depression, or alcoholism), so careful clinical correlation is essential.